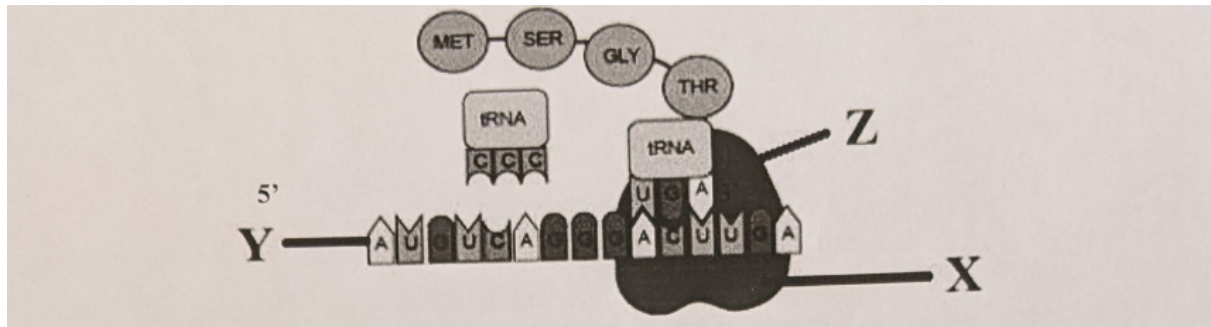


Tutorial_1 Solve



Answers:

1. The process depicted in the diagram is called **Translation**.

Function: It is the process in which **ribosomes synthesize proteins** by reading the mRNA sequence and assembling amino acids into a polypeptide chain.

2. The component X is called **Smaller unit of Ribosome** and its function is **decoding the genetic information**

3. The component Z is called **Larger sub unit of Ribosome**

Description: The **larger subunit of the ribosome** play a vital role in translation and also responsible for **catalyzing peptide bond formation** during protein synthesis. It contains the **peptidyl transferase center (PTC)**, which responsible for the linking of amino acids.

- In **prokaryotes**, the larger subunit is **50S** (composed of **23S rRNA**, **5S rRNA**, and **proteins**).
- In **eukaryotes**, it is **60S** (composed of **28S rRNA**, **5.8S rRNA**, **5S rRNA**, and **proteins**).

It works with the **smaller subunit** to translate mRNA into a polypeptide chain. 🚀

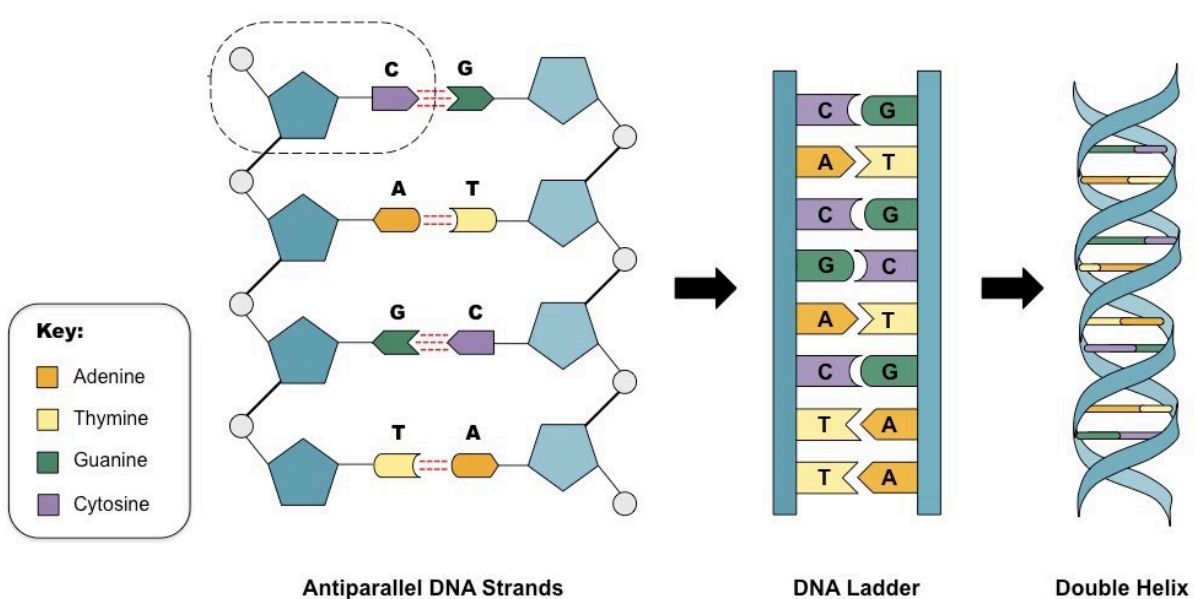
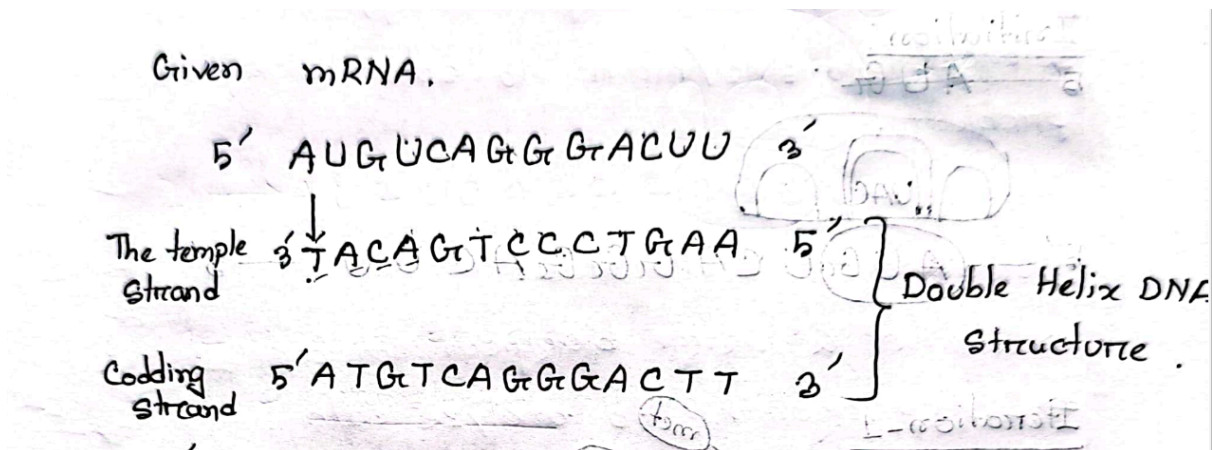
4. Y is called **mRNA (Messenger RNA)**.

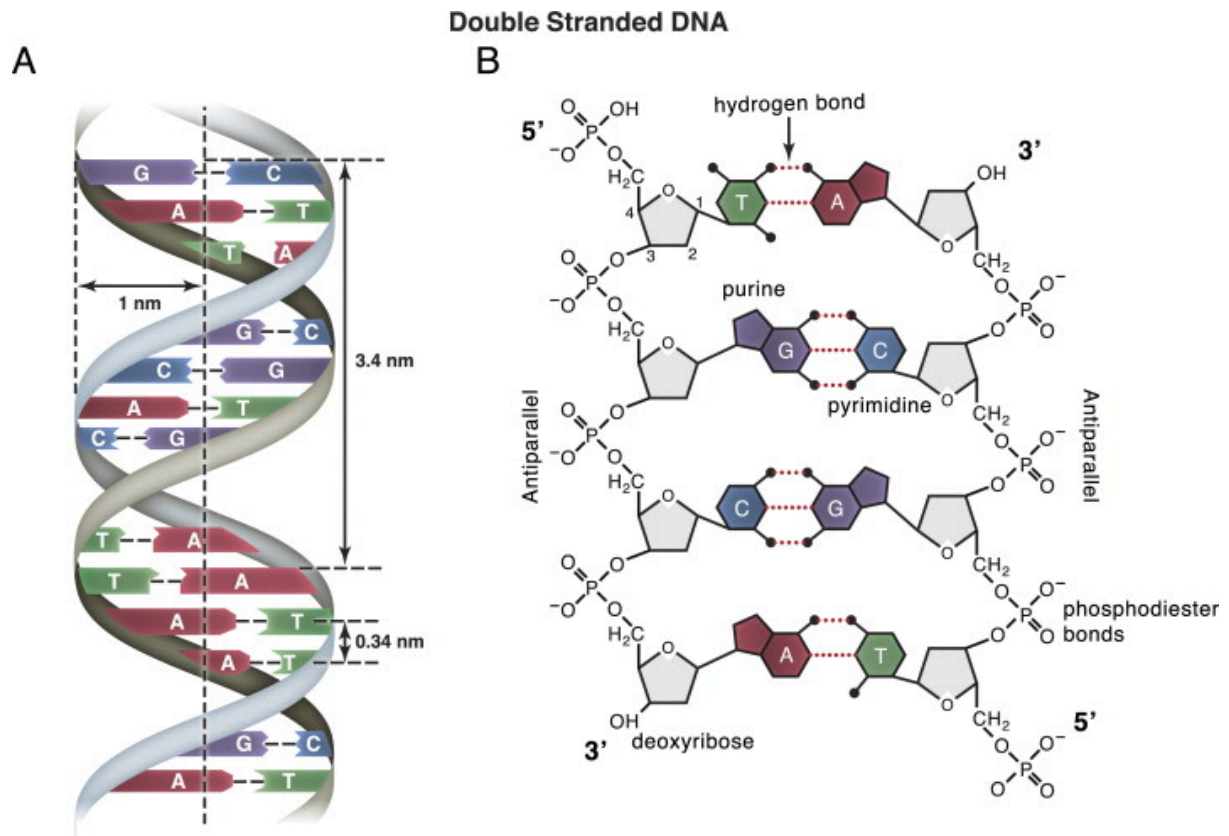
Function: It carries genetic instructions from **DNA** in the nucleus to the ribosome for protein synthesis.

5. Draw the double helix **DNA structure** of the hypothetical gene represented by Y (mRNA).

The answer requires a drawing of a **DNA double-helix** showing complementary base pairing:

- **A (Adenine) pairs with T (Thymine)**
- **C (Cytosine) pairs with G (Guanine)**





6. Illustrate the process identified in (i) for the first two codons.

The first two **codons** from mRNA should be translated into corresponding **tRNA anticodons** and **amino acids** using the genetic code.

- Example:
- If mRNA sequence = **AUG UCU**
- **AUG (Start codon) → tRNA anticodon (UAC) → Methionine (MET)**
- **UCU → tRNA anticodon (AGA) → Serine (SER)**
- The illustration should include ribosome, mRNA, tRNA, and amino acids forming a polypeptide chain.

(vi) Illustrate the process identified in (i) for the first two codons:

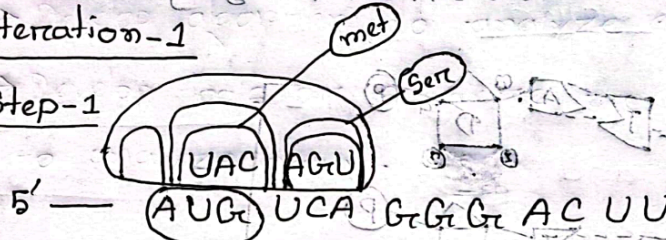
Initiation:

5' — AUG — 3'

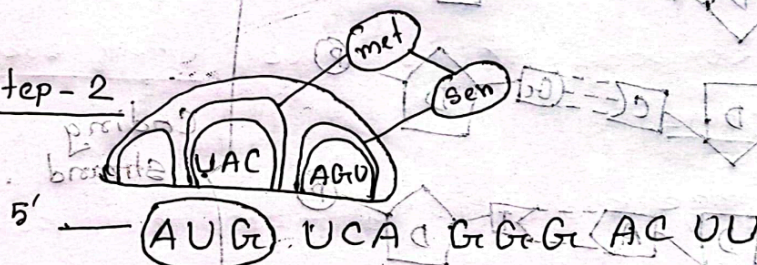


Iteration-1

Step-1



Step-2



Step-3



two codon translation process.

Q2. Consider the following coding strand DNA sequence of the hypothetical color producing gene in some eukaryotic cell of an organism. [use the codon table provided at the end] [5]

	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	
DNA	T	A	C	G	C	T	T	C	G	A	A	G	C	T	C	A	A	C	G	G	C	A	C	T	
mRNA																									
Amino acid sequence																									

After the nitrogenous base at *position 5* being mutated to nitrogenous base A, investigate the impact on the nature of the protein.

Answers:

Q2. DNA to mRNA Transcription and Mutation Impact Analysis

The given DNA coding strand sequence is:

5' - C A A G T C T C A A C G G C A C T - 3'

Step 1: Transcribe DNA to mRNA

The mRNA sequence is transcribed by replacing **T with U**, keeping **A, C, and G the same**:

mRNA: 5' - C A A G U C U C A A C G G C A C U - 3'

Answer to the Q.NO-2

DNA → TAC GCT TCG AAG CTC AAC GGC CACT
mRNA → AUG CGA AGC UUC GAG UUG CCG UGA
 met Arg Ser Phe Glu Leu Pro Stop
 ↓
 start

if the nitrogenous base position 5 being mutated to nitrogenous base A, then the mRNA & protein sequence will change

DNA → TAC GCA TCG AAG CTC AAC GGC CACT
mRNA → AUG CGU AGC UUC GAG UUG CCG UGA
 met Arg Ser Phe Glu Leu Pro Stop
 ↓
 start

Here we see that if the 5th nitrogenous base mutated to A then the codon in that position is changed from CGA to CGU.

and CGA & CGU both represent the same amino acid. So, there shouldn't be any problem.

What is bio_informatics ?



Bioinformatics is an interdisciplinary field that combines **biology, computer science, and mathematics** to analyze and interpret biological data, especially **genomic and proteomic sequences**

Goals of bio_informatics

1. To construct and manage biological database.
2. To develop algorithm for solving different types of biological problem.
3. To develop Tools or softwares or web browser to analyze data.